

Loss of sea ice and albedo could push the arctic towards an ice-free future

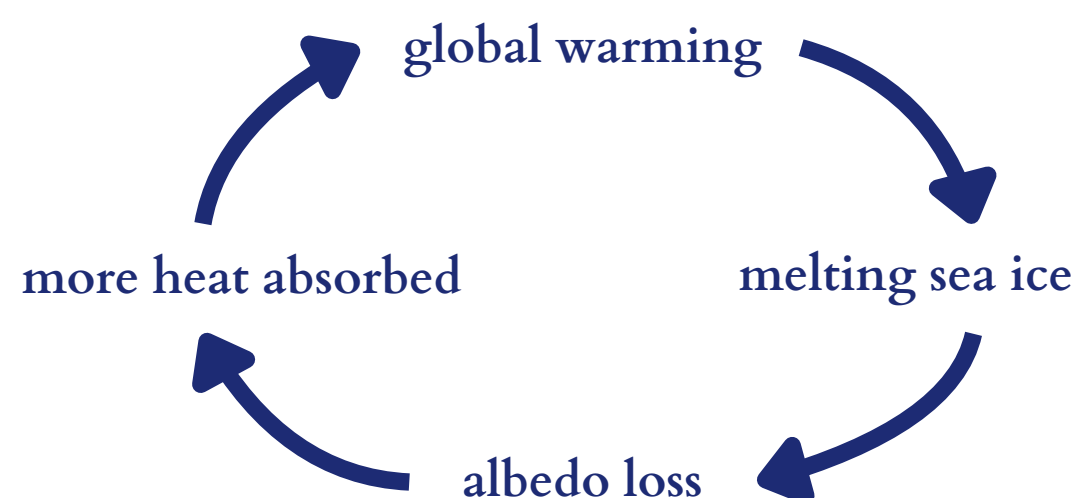
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Project Objective

This project's goal was to explore how climate change is affecting the sea ice extent within the Arctic region of Barents Sea and how it will ultimately affect surface albedo, which is the reflectivity of a surface.

Barents Sea is a key area of concern due to its exposure to arctic amplification, a phenomenon where the Arctic region is currently experiencing warming nearly 4 times the rest of the globe has (Rantanen et al., 2022).

Components of climate change, melting sea ice and albedo all come together to form what is known as the ice-albedo feedback loop, this mechanism is one of the key drivers of arctic amplification



In this positive feedback loop, global warming leads to melting sea ice which means there is a reduction in albedo as sea ice is highly reflective. The sea ice melt exposes the darker ocean underneath that has a low albedo which instead absorbs the incoming radiation from the sun, contributing to self reinforcing nature of the cycle.

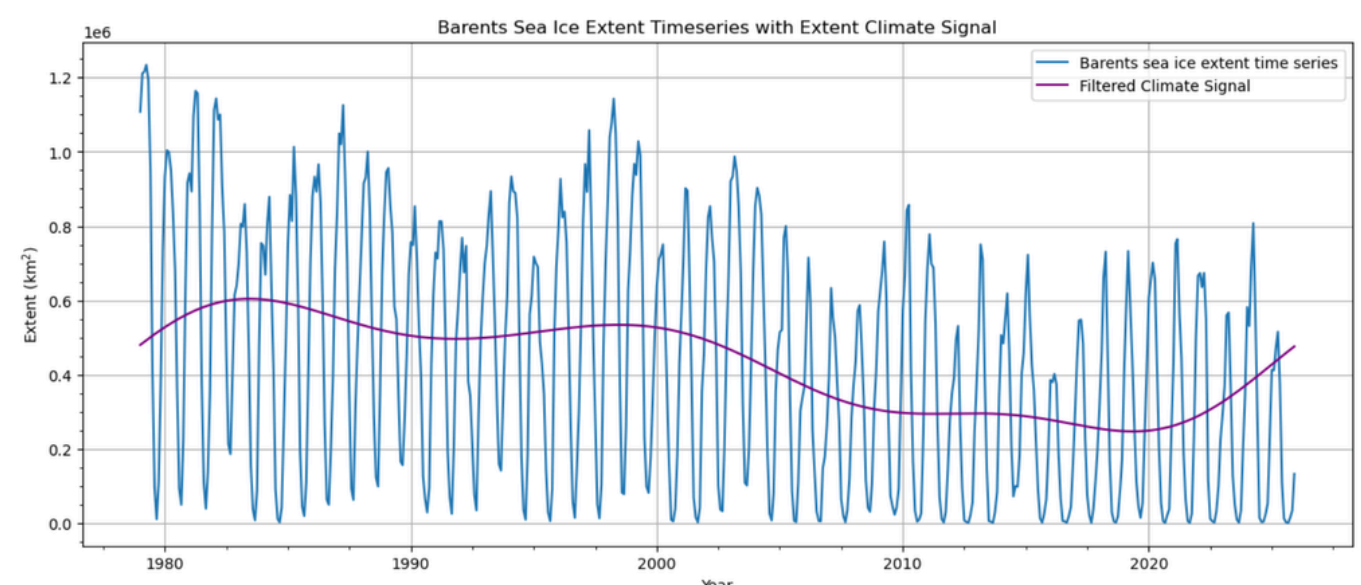
To further explore the implications of this feedback loop, Shared Socioeconomic Pathways obtained from the IPCC AR6 was used to project how continued global warming from CO2 emissions would affect albedo loss.

Datasets used:

- 1) Sea Ice Extent - National Snow and Ice Data Center
- 2) Albedo - Copernicus Data Store ERA5 Reanalysis Monthly averages

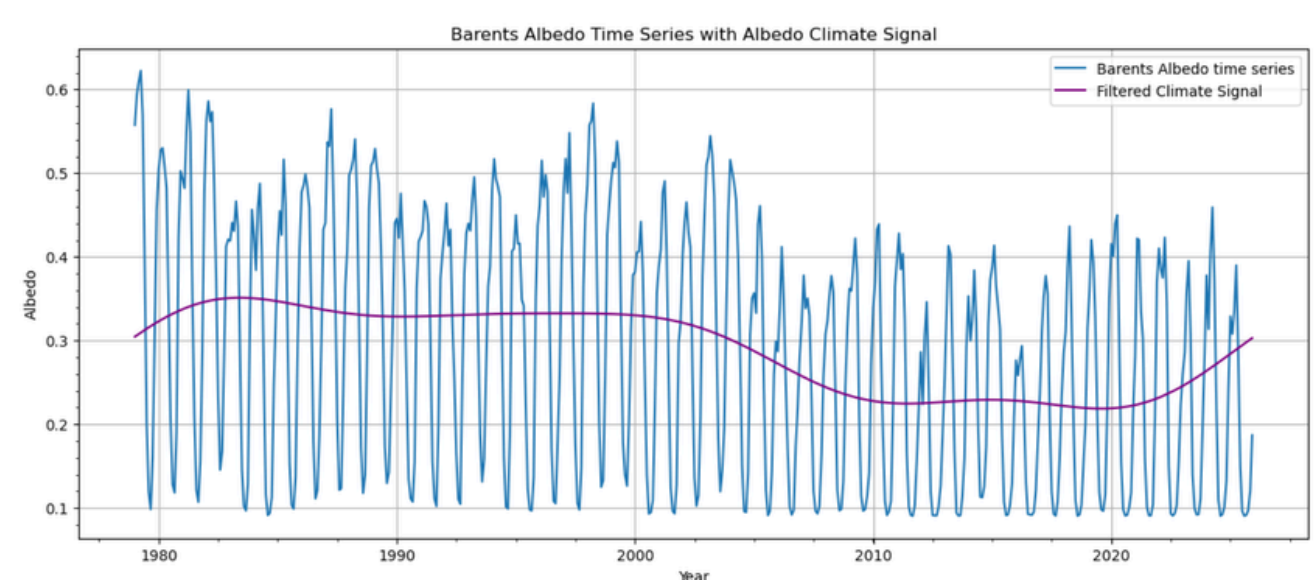
Inverse Fourier transform and signal extraction

Isolated climate signal for sea ice extent



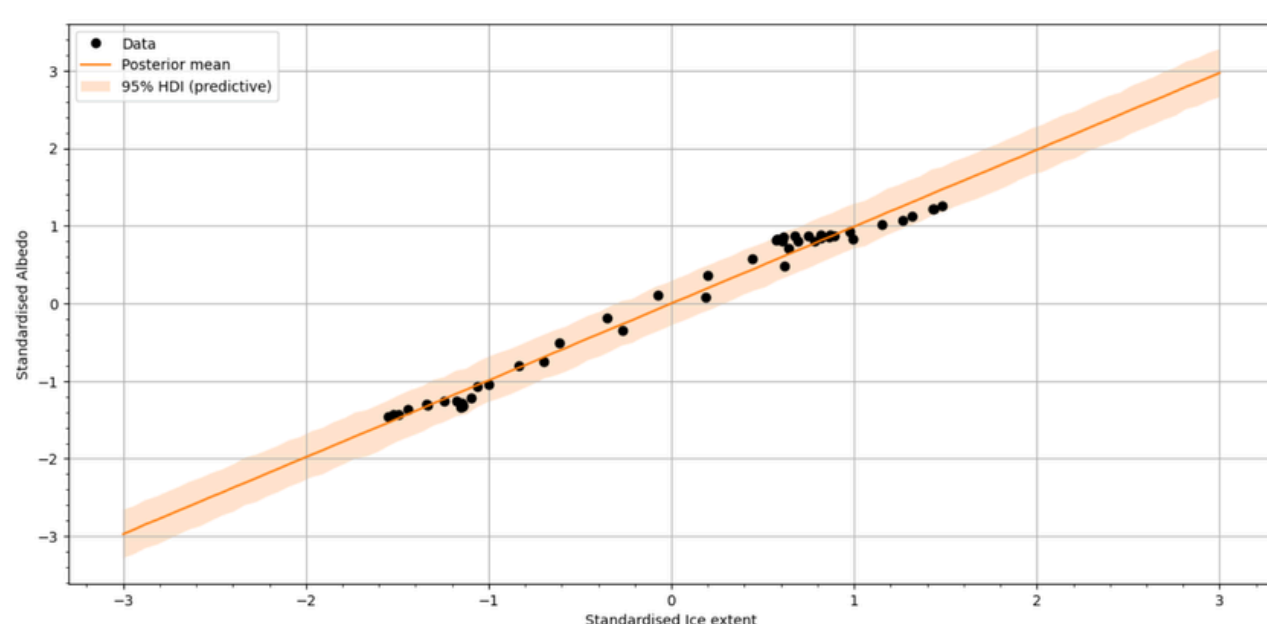
Having filtered cycles that occur more frequently than 10 years The climate signal plotted against the raw data shows a long term decline in Sea ice Extent, consistent with what is currently being seen in the literature

Isolated climate signal for albedo



After filtering out seasonal cycles, the climate signal plotted against the raw albedo data also shows a long term decline in albedo, consistent with what is currently being seen in the literature and also consistent with the sea ice plot.

Linear Bayesian Regression



The bayesian regression quantifies the historical relationship between sea ice extent and albedo.

$$\text{Albedo} = \alpha + \beta * \text{Sea Ice Extent}$$

The regression produced a beta value of 0.99, meaning that the decline and relationship between sea ice and albedo is at at almost 1:1, showing that decline in sea ice is the reason for a decline in albedo

This regression also produced confidence intervals of 0.95-1.00



Numerical simulation

The numerical simulation shows that across the three Shared Socioeconomic Pathways, SSP5-8.5 is projected to have the fastest decline, reaching -3 standard deviations from the 2025 value of 0.3.

By 2100:

SSP5-8.5 (high emissions scenario) estimated to have an albedo of 0.14

SSP2-4.5 (middle of the road) estimated to have an albedo of 0.21

SSP1-2.6 (sustainability) estimated to have an albedo of 0.27

For Reference Albedo spans from 1 (being the highest) to 0 (the lowest). The higher the more reflective the surface

